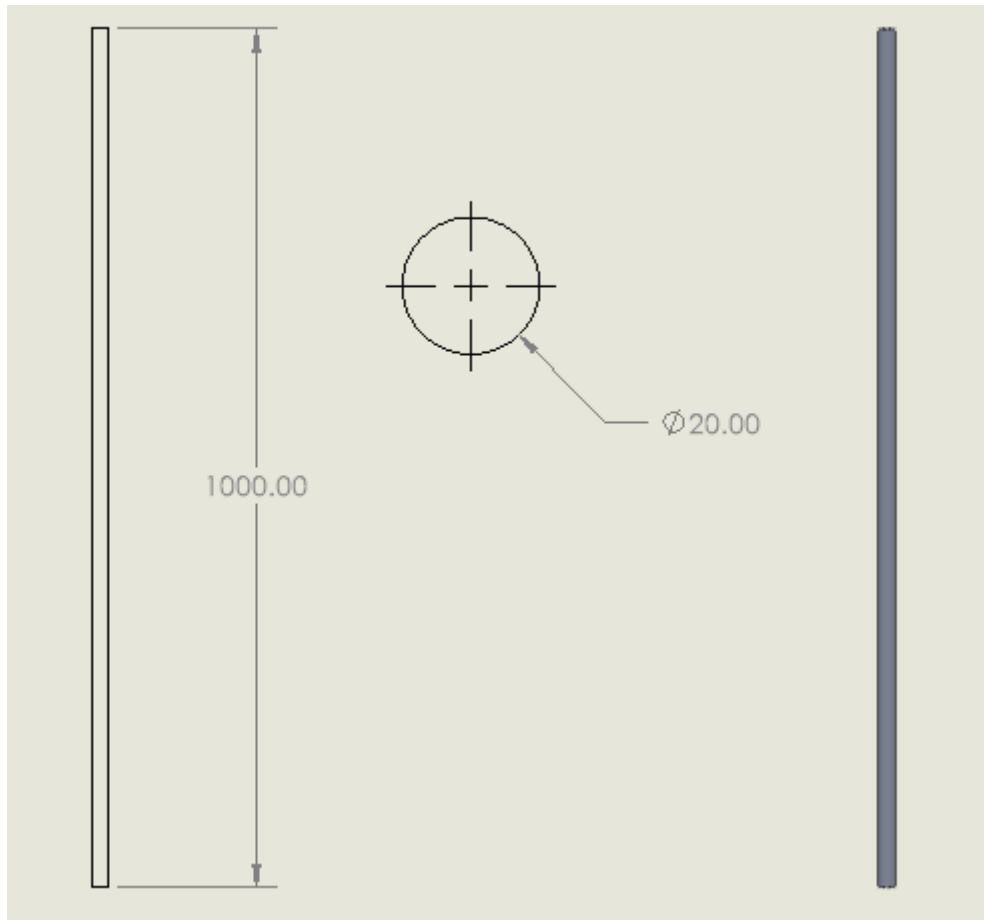


Aim: To determine the critical buckling load of a solid circular column using Eigenvalue Buckling analysis in ANSYS Workbench.

Diagram:



Boundary Conditions:

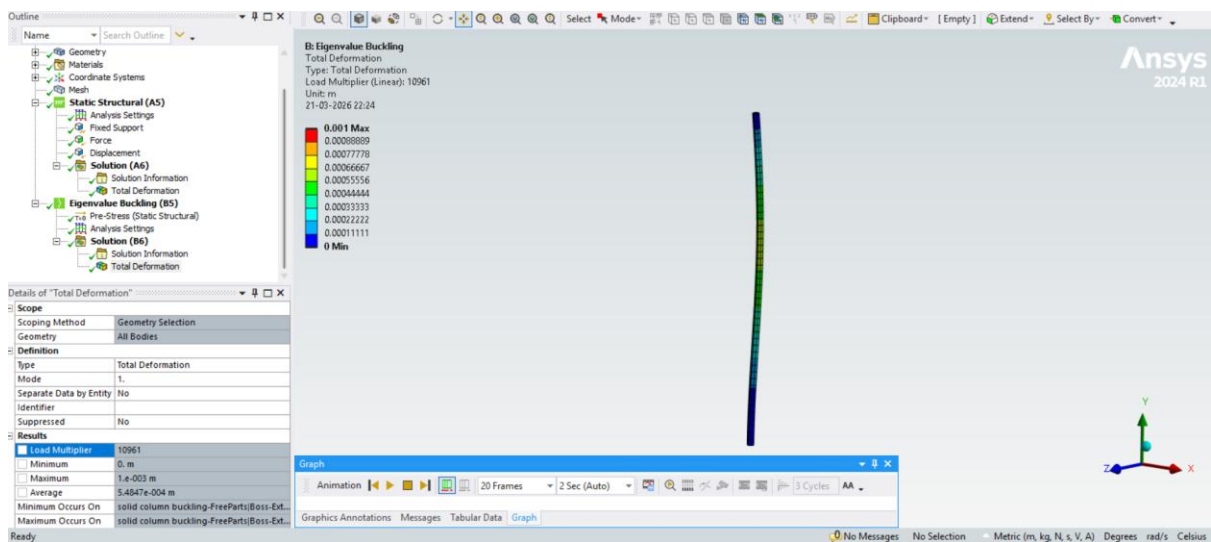
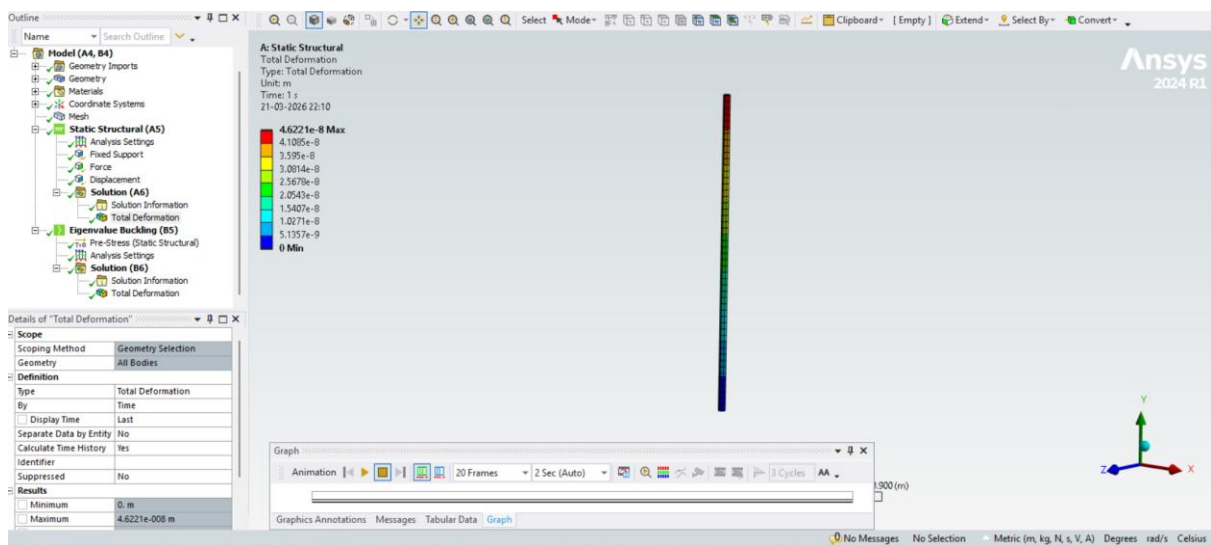
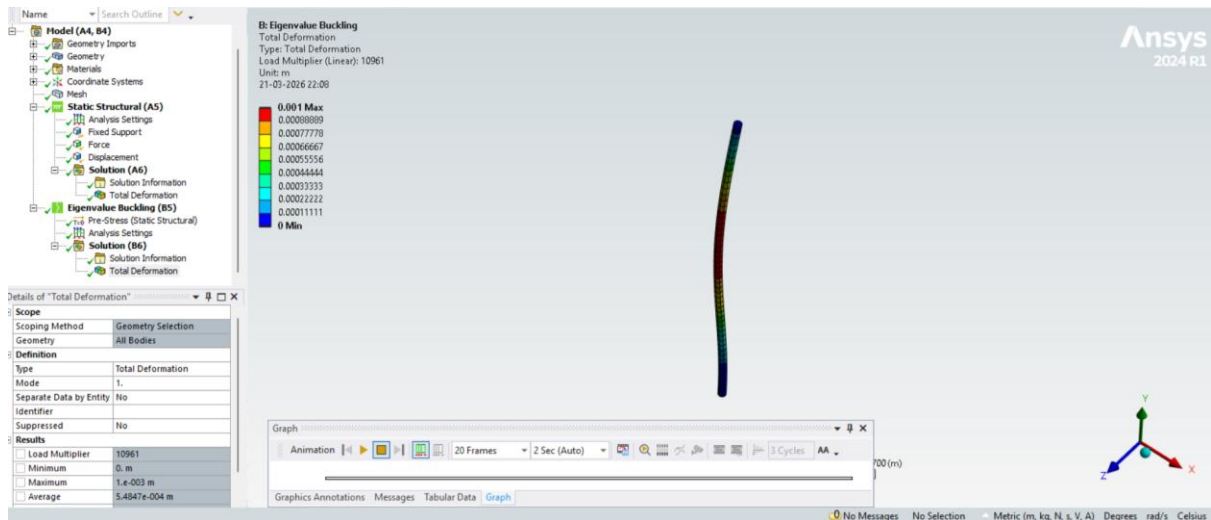
One End: Free

Other End: Displacement- X-axis=0

Y-axis=Free

Z-axis=0

Force= -1N along Y-axis



Details of "Total Deformation" ▾ 🔍 □ ×	
[-] Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
[-] Definition	
Type	Total Deformation
Mode	1.
Separate Data by Entity	No
Identifier	
Suppressed	No
[-] Results	
☐ Load Multiplier	10961
☐ Minimum	0. m
☐ Maximum	1.e-003 m
☐ Average	5.4847e-004 m
Minimum Occurs On	solid column buckling-FreeParts Boss-Ext...
Maximum Occurs On	solid column buckling-FreeParts Boss-Ext...

$E = 69 \text{ GPa}$

Length $L = 1000 \text{ mm}$

Diameter $d = 20 \text{ mm}$

End condition: Fixed–Pinned $\rightarrow K = 0.7$

ANSYS RESULT:

$P_{\text{(critical)}} = \text{load multiplier} \times \text{force} = 10961 \times 1 = 10961 \text{ N}$

THEORETICAL RESULT:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2} = 10915.5 \text{ N}$$

ERROR: 0.4% which is less than 1%